You'll work with LIKE queries and develop a similarity search solution using GIN indexes and trigrams (See: *Postgres GIN Index in Rails*[9]). Trigrams can be captured and a GIN index can be used to accept a user's search input and find similar text. Trigrams will help create matches on similarity. The query planner will use a bitmap index scan with this type of GIN index.

Run the following query from psql that performs an ILIKE query for some rows in the users table. The table should be populated from the data generators and have at least 20 thousand rows in it. If you wish to reset the Rideshare database, run bin/rails db:reset and then bin/rails data_generators:generate_all from your Terminal.

sql/query_ilike.sql

```
SELECT DISTINCT first_name FROM users WHERE first_name ILIKE 'henr%';
```

ILIKE is being used for case insensitive similarity searching. Lowercase "henr" is used as the partial matching text. Passing the text "henr" with case insensitivity matches on Henrietta, Henriette, and Henry.

You should get those three results when you run the query above. The query performed a Sequential Scan of the whole table. You can speed that up using a GIN index.

You'll enable an extension and add the index using a Active Record Migration.

The following Migration has not been added to Rideshare. To add it, generate a migration file by running this command from your terminal:

```
bin/rails g migration AddTempUsersGinTrgmOpsIndex
```

In the generated Migration file, copy and paste the implementation body from the example below. The Migration will enable the extension and then add the index.

ruby/migration_similarity_search.rb

```
class AddFirstNameSimilaritySearch < ActiveRecord::Migration[7.0]
  disable_ddl_transaction!

  def change
    enable_extension('pg_trgm')

    add_index(:users,
              :first_name,
              name: 'users_first_name_gin_trgm_ops_idx',
              using: 'gin',
              opclass: :gin_trgm_ops,
              algorithm: :concurrently)
```

9. https://blog.kiprosh.com/postgres-gin-index-in-rails/